

Social Media Platform Case Study Assignment – MongoDB Replication & Replica Sets

Case Study Title

Designing a Scalable Social Media Analytics Platform using MongoDB Replica Sets

Scenario

A social media company similar to Instagram or Facebook wants to build a backend system to manage:

- User profiles
- Posts and feeds
- Likes and comments
- Followers and following
- User activity logs
- Trending hashtag analytics

The platform serves millions of active users daily. Since users continuously create posts, like content, and interact in real time, the company requires:

- High availability
- Fault tolerance
- Data redundancy
- Read scalability
- Continuous monitoring

To achieve this, the engineering team decides to implement **MongoDB Replication using Replica Sets**.

Assignment Objective

Students must design and implement a beginner-friendly MongoDB-based social media analytics system that demonstrates:

- Replica Set Architecture
- Primary-Secondary Replication
- High Availability Concepts
- Replication Monitoring
- CRUD Operations using MongoDB

The assignment focuses on understanding MongoDB replication concepts practically through a social media use case.

Business Requirements

The social media platform should support the following modules:

1. User Profile Management

Store user details such as:

- Username
- Email
- Bio
- Followers count
- Following count
- Account creation date

Example Features

- Create profile
- Update bio
- View profiles
- Delete inactive users

2. Posts and Feed Management

Store:

- User posts
- Captions
- Hashtags
- Post timestamps
- Likes count

Example Features

- Create posts
- View feeds
- Update captions
- Delete posts

3. Likes and Comments Tracking

Track:

- Likes on posts
- Comments
- Comment timestamps
- User interactions

Example Features

- Add likes
- Add comments
- View engagement analytics

4. Followers and Following System

Store:

- Followers list
- Following list
- Follow timestamps

Example Features

- Follow users
- Unfollow users
- View followers

5. Activity Logging

Track:

- Login activity
- Post creation
- Like activity
- Comment activity
- Device activity

Example Features

- Security monitoring
- User engagement analysis
- Device tracking

Beginner-Friendly Folder Structure

```
mongodb-replication-socialmedia/  
├── data/  
│   ├── db1/  
│   ├── db2/  
│   └── db3/  
├── scripts/  
│   └── setupReplica.js
```

—	insertUsers.js
—	insertPosts.js
—	insertComments.js
—	insertFollowers.js
—	insertActivityLogs.js
—	checkData.js
—	updateData.js
—	deleteData.js
—	failoverDemo.js

Collections and Columns

1. users Collection

users

Column	Type	Description
userId	String	Unique user ID
username	String	Username
email	String	User email
bio	String	Profile bio
followersCount	Number	Total followers
followingCount	Number	Total following

2. posts Collection

posts

Column	Type	Description
postId	String	Post ID
userId	String	Linked user
caption	String	Post caption
hashtags	Array	Tags used
likesCount	Number	Total likes
createdAt	Date	Post time

3. comments Collection

comments

Column	Type	Description
commentId	String	Comment ID
postId	String	Linked post
userId	String	Comment user
commentText	String	Comment
commentedAt	Date	Comment timestamp

4. followers Collection

followers

Column	Type	Description
followId	String	Follow ID
followerId	String	Follower user
followingId	String	Following user
followedAt	Date	Follow timestamp

5. activity_logs Collection

activity_logs

Column	Type	Description
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activityId	String	Activity ID
userId	String	User reference
activityType	String	Login/Post/Like
device	String	Mobile/Laptop
activityTime	Date	Activity timestamp

CRUD Operations Requirements

CREATE Operations

Students should create scripts for:

Script	Operation
insertUsers.js	Insert users
insertPosts.js	Insert posts
insertComments.js	Insert comments
insertFollowers.js	Insert followers
insertActivityLogs.js	Insert logs

READ Operations

Using:

`checkData.js`

Students should:

- View users
- View posts
- View comments
- View followers
- View activity logs

UPDATE Operations

Using:

`updateData.js`

Students should:

- Update bio
- Update captions
- Update likes count

DELETE Operations

Using:

`deleteData.js`

Students should:

- Delete inactive users
- Delete unwanted comments
- Delete posts

Replica Set Operations

setupReplica.js

Students should:

- Initialize replica set
- Configure db1, db2, db3
- Set Primary and Secondary nodes

failoverDemo.js

Students should:

- Stop Primary node
- Observe automatic election
- Verify new Primary node